$$J(\theta) = \underset{\tau\sim\pi_\theta}{E}[R(\tau)] = \int R(\tau)\cdot\pi_\theta(\tau)\,d\tau \quad \left[\theta = (\theta_1, \theta_2)\right]$$

$$\nabla J(\theta) = \int R(\tau)\cdot\nabla\pi_\theta(\tau)\,d\tau = (*)$$

$$\left[(\log(f))' = \frac{1}{f}\cdot f'/\cdot f \;\Big\{\; f' = f\cdot(\log(f))'\right] \qquad \nabla\pi_\theta(\tau) = \pi_\theta(\tau)\cdot\nabla\log(\pi_\theta(\tau))$$

$$(*) = \int R(\tau)\cdot\pi_\theta(\tau)\cdot\nabla\log(\pi_\theta(\tau))\cdot d\tau =$$

$$= \int \underbrace{R(\tau)\cdot\nabla\log(\pi_\theta(\tau))}\cdot\pi_\theta(\tau)\,d\tau = \cancel{E[R(\tau)]}$$

$$= \underset{\tau\sim\pi_\theta}{E}[R(\tau)\cdot\nabla\log(\pi_\theta(\tau))] = (2)$$

$$\pi_\theta(\tau) = P(s_0)\cdot\pi_\theta(a_0|s_0)\cdot P(s_0,a_0,s_1)\cdot\pi_\theta(a_1|s_1)\cdot P(s_1,a_1,s_2)\ldots$$

$$\log(\pi_\theta(\tau)) = \log(P(s_0)) + \log\pi_\theta(a_0|s_0) + \log(P(s_0,a_0,s_1)) + \ldots$$

$$\nabla\log(\pi_\theta(\tau)) = \cancel{\nabla\log(P(s_0))}^{\,0} + \nabla\log(\pi_\theta(a_0|s_0)) + \cancel{\nabla\log(P(s_0,a_0,s_1))}^{\,0} + \ldots =$$

$$= \sum_t \nabla\log(\pi_\theta(a_t|s_t))$$

$$\Rightarrow \boxed{\nabla\log(\pi_\theta(\tau)) = \sum_t \nabla\log(\pi_\theta(a_t|s_t))}$$

$$(2) = \underset{\tau\sim\pi_\theta}{E}\left[R(\tau)\cdot\sum_t \nabla\log(\pi_\theta(a_t|s_t))\right] =$$

$$= E\left[\sum_t \nabla\log(\pi_\theta(a_t|s_t))\cdot R(\tau)\right] =$$

$$= E\left[\sum_t \nabla\log(\pi_\theta(a_t|s_t))\cdot G_t\right] \approx$$

$$\approx \frac{1}{M}\left[\sum_{i=1}^{M}\sum_t \nabla\log(\pi_\theta(a_t|s_t))\cdot G_t\right]$$

$$\Rightarrow \boxed{\nabla J(\theta) \approx \frac{1}{M}\left[\sum_{i=1}^{M}\cdot\sum_t \nabla\log(\pi_\theta(a_t|s_t))\cdot G_t\right]}$$

$$\nabla \log(\pi_\theta(a_t|s_t)) = \left(\frac{d\log(\pi_\theta(a_t|s_t)}{d\theta_1}, \frac{d\log(\pi_\theta(a_t|s_t))}{d\theta_2}\right) = ? \quad (3)$$

$\pi_\theta(a_t|s_t)$ - gustina vjerovatnoće u tački a

~~$\pi_\theta(a_t|s_t) = \frac{1}{2\sigma^2}$~~

$$\pi_\theta(a_t|s_t) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(a-\mu)^2}{2\sigma^2}}$$

$$\log(\pi_\theta(a_t|s_t)) = \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) + \log\left(e^{-\frac{(a-\mu)^2}{2\sigma^2}}\right) =$$

$$= \cancel{\log 1}^{0} - \cancel{\log(\sigma\sqrt{2\pi})}^{0} - \frac{(a-\mu)^2}{2\sigma^2} = -\frac{(a-\mu)^2}{2\sigma^2}$$

$$\Rightarrow \boxed{\log(\pi_\theta(a_t|s_t)) = -\frac{(a-\mu)^2}{2\sigma^2}}$$

$$\frac{d\log(\pi_\theta(a_t|s_t))}{d\mu} = -\frac{1}{2\sigma^2} \cdot 2(a-\mu)(-1) = \frac{a-\mu}{\sigma^2}$$

$$\frac{d\mu}{dz} = \frac{d}{dz}(\tanh(z)) = 1 - (\tanh(z))^2 = 1 - \mu^2$$

$$\frac{dz}{d\theta_1} = \frac{d}{d\theta_1}(x_1\theta_1 + x_2\theta_2 + b) = x_1$$

$$\Rightarrow \frac{d\log(\pi_\theta(a_t|s_t))}{d\theta_1} = \frac{d\log(\pi_\theta(a_t|s_t))}{d\mu} \cdot \frac{d\mu}{dz} \cdot \frac{dz}{d\theta_1} =$$

$$= \cancel{\frac{(a-\mu)^2}{\sigma^2}} \cdot \cancel{2}(1-\mu^2) \cdot x_1 = -\frac{(a-\mu)^2}{\sigma^2}(1-\mu^2) \cdot x_1$$

$$\Rightarrow \frac{d\log(\pi_\theta(a_t|s_t))}{d\theta_2} = \cancel{\frac{(a-\mu)^2}{\sigma^2}} \cdot (1-\mu^2) \cdot x_2$$

Neka je $\Delta = \cancel{\frac{(a-\mu)^2}{\sigma^2}} \cdot (1-\mu^2)$.

$\Longrightarrow$

$$\frac{d \log(\pi_\theta(a_t|s_t))}{d\theta_1} = \Delta \cdot x_1$$

$$\frac{d \log(\pi_\theta(a_t|s_t))}{d\theta_2} = \Delta \cdot x_2, \text{ gdje je } \Delta = \frac{(a-\mu)}{\sigma^2} \cdot (1-\mu^2)$$

$$\frac{dJ}{d\theta_1} \approx \frac{1}{N}\left[\sum_{i=1}^{n} \cdot \sum_{t} G_t \cdot \Delta_t \cdot x_{1_t}\right] =$$

$$= \frac{1}{N}\left[\sum_{i=1}^{n} \cdot \sum_{t} \delta_t \cdot x_{1_t}\right]$$

$$\Rightarrow \left| \frac{dJ}{d\theta_1} \approx \frac{1}{N}\left[\sum_{i=1}^{n} \sum_{t} \delta_t \cdot x_{1_t}\right] \right.$$

$$\left. \frac{dJ}{d\theta_2} \approx \frac{1}{N}\left[\sum_{i=1}^{n} \sum_{t} \delta_t \cdot x_{2_t}\right], \right.$$

$$\text{gdje je } \delta_t = \Delta_t \cdot G_t = \frac{a-\mu}{\sigma^2}(1-\mu^2) \cdot G_t$$

$$\Rightarrow \left\{ \begin{array}{l} \theta_1 = \theta_1 + \alpha \cdot \frac{dJ}{d\theta_1} \\ \theta_2 = \theta_2 + \alpha \cdot \frac{dJ}{d\theta_2} \end{array} \right\} \boxed{\theta = \theta + \alpha \cdot \nabla J(\theta)}$$